

Lakshya Aryan

Portfolio

9897983015 | [Gmail](#) | [LinkedIn](#) | [GitHub](#)

EDUCATION

Indian Institute Of Technology Varanasi (BHU)

Bachelor of Technology

Varanasi, UP

2022 – 2026

EXPERIENCE

Quantitative Researcher Internship

Jun 2025 – Sep 2025

AR Financial Securities

Remote

- Enhanced option pricing accuracy using the Variance Gamma model beyond Black-Scholes assumptions.
- Processed data for 2,500+ stocks and built analytical dashboards for market research and strategy evaluation.
- Developed a KAMA-based portfolio construction strategy across diversified equity universes.

Quantitative ML Developer Internship

May 2025 – Jun 2025

SPRINGPAD WEALTH

Remote

- Built Random Forest and XGBoost models to forecast Nifty and BankNifty market movements.
- Designed and backtested a Z-score mean-reversion strategy on Gold-Silver pairs using five years of data.
- Achieved 54.06% returns with a 0.67 Sharpe ratio while limiting maximum drawdown to 16.86%.

PROJECTS

Low-Latency Trade Matching Engine | *C++, STL, Multithreading*

- Built a low-latency exchange matching engine supporting FIFO and price-time priority order execution.
- Implemented cache-efficient order book structures using Red-Black Trees, hash maps, and linked lists.
- Optimized critical trading operations to $O(1)$ order cancel/execute and $O(\log M)$ matching complexity.

RL-Powered Algorithmic Trading Platform | *Python, React, FastAPI, Stable-Baselines3, Claude AI*

- Designed a DRL-based trading platform supporting PPO, DQN, and A2C strategies for Indian equities.
- Integrated custom trading environments, risk controls, real-time data streaming using technical indicators.
- Built an interactive dashboard with LLM-powered trade rationale generation with paper trading support.

Volatility Index Estimation Calculator | *Python, NumPy, Pandas*

- Built a VIX calculator implementing the CBOE volatility index methodology from options market data.
- Developed strike selection, OTM option filtering, and variance-swap based volatility estimation models.
- Automated forward price computation and variance aggregation across expiries for robust VIX estimation.

TECHNICAL SKILLS

- Programming:** C, C++, Python, SQL, Shell
- Quant Finance:** Options Pricing, Volatility Modeling, Backtesting, Algorithm Trading
- Tools:** Probability, Statistics, Optimization, Numpy, Pandas, Git, Scikit-learn, API

HONOURS & ACHIEVEMENTS

- Star Quant of the Month* – for exceptional performance during internship at AR Financial.
- Certificate of Excellence in Machine Learning & Summer Analytics Program, *IIT Guwahati*.
- National Runner-up, Quantalytics *IIT Jodhpur* – built and backtested a quant strategy.
- Achieved *1744 max LeetCode rating*, demonstrating strong problem-solving skills.

POSITIONS OF RESPONSIBILITY

- Operations Manager, *Technex'24* – managed event operations and cross-team coordination.
- Marketing Executive, *Kashiyatra'23* – led sponsorship outreach and partnership initiatives.